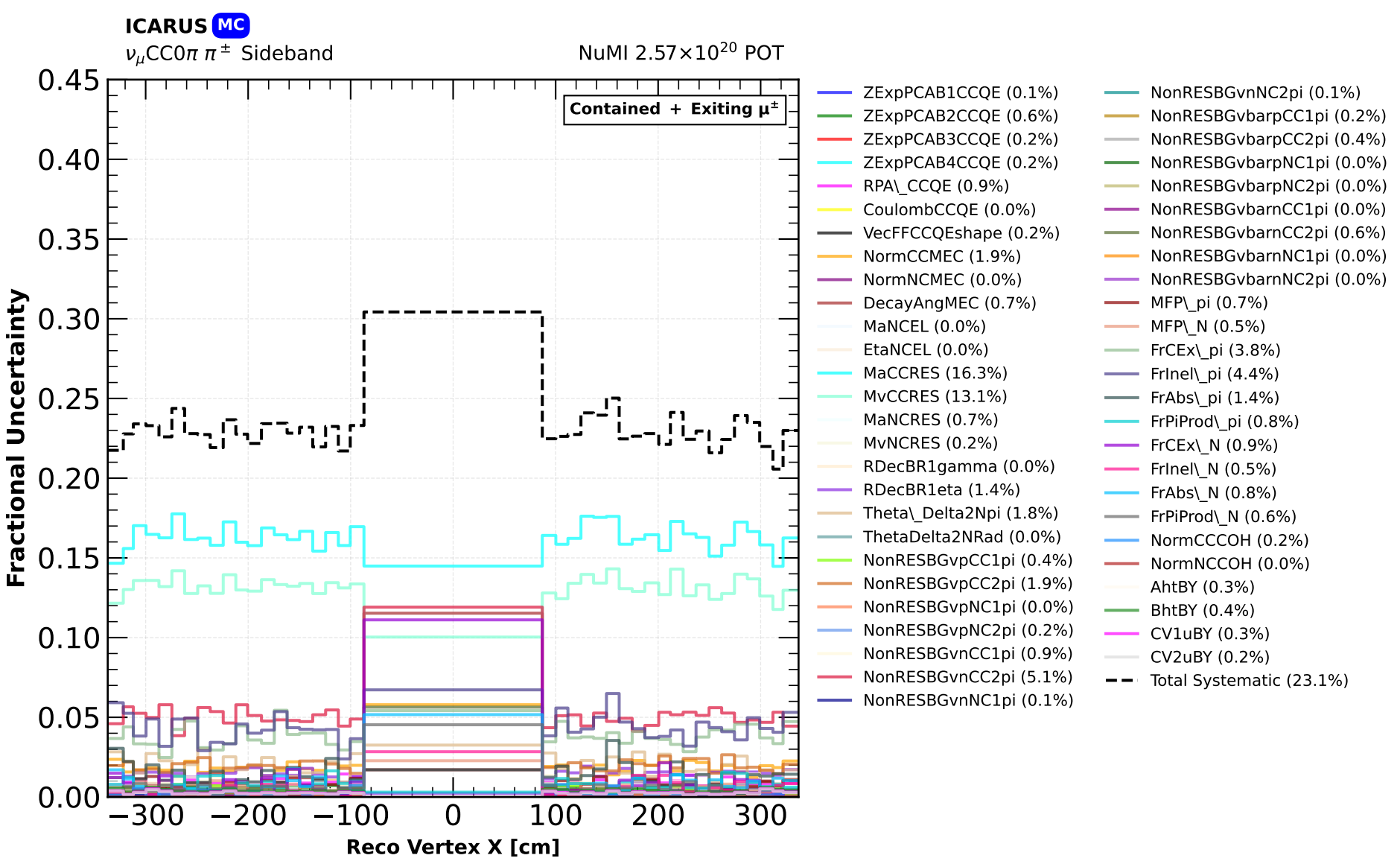
ICARUS MC
$\nu_\mu$CC0$\pi$ $\pi^\pm$ Sideband
NuMI 2.57×10$^{20}$ POT
Contained + Exiting μ$^\pm$
Fractional Uncertainty
Reco Vertex X [cm]
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
−300
−200
−100
0
100
200
300
ZExpPCAB1CCQE (0.1%)
ZExpPCAB2CCQE (0.6%)
ZExpPCAB3CCQE (0.2%)
ZExpPCAB4CCQE (0.2%)
RPA_CCQE (0.9%)
CoulombCCQE (0.0%)
VecFFCCQEshape (0.2%)
NormCCMEC (1.9%)
NormNCMEC (0.0%)
DecayAngMEC (0.7%)
MaNCEL (0.0%)
EtaNCEL (0.0%)
MaCCRES (16.3%)
MvCCRES (13.1%)
MaNCRES (0.7%)
MvNCRES (0.2%)
RDecBR1gamma (0.0%)
RDecBR1eta (1.4%)
Theta_Delta2Npi (1.8%)
ThetaDelta2NRad (0.0%)
NonRESBGvpCC1pi (0.4%)
NonRESBGvpCC2pi (1.9%)
NonRESBGvpNC1pi (0.0%)
NonRESBGvpNC2pi (0.2%)
NonRESBGvnCC1pi (0.9%)
NonRESBGvnCC2pi (5.1%)
NonRESBGvnNC1pi (0.1%)
NonRESBGvnNC2pi (0.1%)
NonRESBGvbarpCC1pi (0.2%)
NonRESBGvbarpCC2pi (0.4%)
NonRESBGvbarpNC1pi (0.0%)
NonRESBGvbarpNC2pi (0.0%)
NonRESBGvbarnCC1pi (0.0%)
NonRESBGvbarnCC2pi (0.6%)
NonRESBGvbarnNC1pi (0.0%)
NonRESBGvbarnNC2pi (0.0%)
MFP_pi (0.7%)
MFP_N (0.5%)
FrCEx_pi (3.8%)
FrInel_pi (4.4%)
FrAbs_pi (1.4%)
FrPiProd_pi (0.8%)
FrCEx_N (0.9%)
FrInel_N (0.5%)
FrAbs_N (0.8%)
FrPiProd_N (0.6%)
NormCCCOH (0.2%)
NormNCCOH (0.0%)
AhtBY (0.3%)
BhtBY (0.4%)
CV1uBY (0.3%)
CV2uBY (0.2%)
Total Systematic (23.1%)